

Ailun (Allen) Shen

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EDUCATION

University of Washington, Paul G. Allen School of Computer Science & Engineering

Seattle, WA

B.S. in Computer Science — Entering Fall 2026 with junior standing • Expected June 2029

RESEARCH INTERESTS

Interpretability and training dynamics of large language models, including how capabilities emerge and are lost over the course of training and which internal representations a model can report on; efficient inference and decoding; simulation-based inference and likelihood-free parameter estimation for scientific data.

RESEARCH EXPERIENCE

Undergraduate Researcher — Department of Physics, University of Washington

Sept 2025 – Present

Under Prof. Shih-Chieh Hsu

- Built a Neural Simulation-Based Inference head on the EveNet foundation model to measure off-shell Higgs signal strength in $gg \rightarrow (h^* \rightarrow) ZZ \rightarrow 4\ell$, benchmarking frozen and fine-tuned backbones against a CARL MLP baseline on $\sim 10M$ events and recovering true μ within 1σ .
- Developing a weakly supervised domain-adaptation method that calibrates latent-event reconstruction under systematic shift, interpolating classifier-based posterior surrogates trained on three discrete systematic templates into a continuous per-event reward so that mass-sensitive observables in dileptonic top-quark pair events remain unbiased.

Undergraduate Researcher — UW Math AI Lab (PermuFormer 2)

Oct 2026 – Present

- Selected for the PermuFormer 2 project, building on PermuFormer, a transformer pretrained on permutation tasks in algebraic combinatorics.

Researcher — Algoverse AI Research

June 2026 – Sept 2026

- Extended the Jacobian-lens framework for verbalizable global workspaces (Gurnee et al., 2026) along the training-time and model-size axes, fitting lenses across six Pythia models (70M to 6.9B) at 14 pretraining checkpoints.
- Found that model size controls whether a verbalizable readout forms but not when it forms, and that the workspace band's geometric signature tracks model size rather than readout formation.
- Accepted as **co-first author** at five NeurIPS 2026 workshops: NeurReps, Interp4Discovery, LP4FM, NewInML, and PriGM.

Independent Researcher — Gradient Descent Emerges via Natural Selection

July 2024 – Aug 2025

- Designed a genetic algorithm to test the hypothesis that evolution can give rise to gradient-like learning in biological brains.
- Showed that evolutionary processes can produce gradient approximators sufficient for neural-network learning.
- Published as **sole author**, IEEE Congress on Evolutionary Computation (CEC) 2025, Hangzhou, China.

Independent Researcher — Space-Time Curvature-Based Data Clustering

Sept 2023 – June 2024

- Reformulated data clustering using principles from General Relativity, achieving accuracy superior to k-means and hierarchical clustering.
- Introduced the Curvature Density (CD) score as a novel clustering metric.
- Published as **first author**, IEEE AMLDS 2025, Tokyo, Japan; awarded 1st Place (Top 2), Software Category, WSSEF 2024.

PUBLICATIONS

Conference Papers

1. A. Shen, "Gradient Descent Emerges via Natural Selection," *IEEE Congress on Evolutionary Computation (CEC)*, June 2025, Hangzhou, China. (Sole author)
2. A. Shen et al., "A Breadth-First Search Algorithm for Data Clustering based on Space-time Curvature," *IEEE AMLDS*, July 2025, Tokyo, Japan. (First author)

3. A. Shen et al., "Pervasive Seizure Detection via One-Class EEG Modeling with Visual Insights," *IEEE SITA*, October 2025. (Second author)

Workshop Papers

4. A. Dabir*, A. Shen* et al., "Verbalizable Representations Emerge before Workspace Functions in Language Models," *NeurIPS 2026 Workshops: NeurReps, Interp4Discovery, LP4FM, NewInML, PriGM*. (Co-first author; *equal contribution)

PRESENTATIONS

- Accepted Poster Presenter, NSF IAIFI Summer Workshop, Harvard University, Aug 2025.

HONORS & AWARDS

- Virtual participant, NSF IAIFI Summer School (2026).
- USAAIO (USA AI Olympiad) Bronze Medal & Student Ambassador (2025).
- USAAIO Round 1 High Honor Roll (2026).
- AIME Qualifier (2022).
- WSSEF 1st Place, Software Category (2024).

PROGRAMS & TRAINING

- Y Combinator Startup School, San Francisco (July 2026).

TEACHING & OUTREACH

Founder & Vice President, Interlake AI Club

- Founded and lead a student AI club; teach AI foundations, run USAAIO preparation, and organize community AI projects.

LEADERSHIP & SERVICE

Co-Founder, PingPongParkinson WA Chapter

- Co-founded a table-tennis-therapy nonprofit serving patients with Parkinson's disease.

SKILLS

Programming: Python, Java

ML / Deep Learning: PyTorch, Hugging Face (Transformers & Datasets), scikit-learn

Scientific Computing: NumPy, pandas, Matplotlib

Cloud: Google Cloud Platform (GCP), Amazon Web Services (AWS), Microsoft Azure

Tools: Git, Weights & Biases

ACTIVITIES

- **Table Tennis:** Team Captain; WA State Junior Boys Runner-Up (2024); U.S. Nationals Bronze Medal, U1600 Junior (2024).
- **Music:** WMEA All-State Choir, Bass 2 and Bass Section Leader (2026); 1st Place, WorldStrides Music Festival, Anaheim (2024) and San Francisco (2026).